

JEFFREY TSENG

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Purdue University

PROFESSIONAL SUMMARY

Full-Stack Engineer with professional experience architecting **high-fidelity React and Framer frontends**, scalable **RAG-based AI backends**, and **React Native mobile applications**.

EDUCATION

Purdue University, West Lafayette, IN Expected Graduation: May 2028
Bachelor of Science in Computer Science and Mathematics GPA: 4.0/4.0
Relevant Classes: CS390: Web Applications Programming, CS193: Computer Science Tools, CS180: Problem Solving & Object-Oriented Programming, MA261: Multivariable Calculus

RELEVANT EXPERIENCE

Google Developer Groups Purdue University
Technical Teaching Assistant August 2025 – Present

- Designed and deployed take-home projects for **300+ students teaching React, Google Cloud, Vertex AI, & Tensorflow**
- **Mentored 150+** developer role students, providing technical assistance on **Flutter, Firebase, and full-stack projects**
- **Selected as one of two technical mentors** from over 250+ applicants

Hack the Future Purdue University
Team Lead October 2025 – Present

- **Led cross-functional development teams** of 10+ student engineers to design and deploy full-stack applications for local nonprofits, integrating **React, Node.js/FastAPI, and MongoDB**
- Conducted **code reviews** and managed teams with **Git workflow**, fostering a **collaborative development environment**
- Collaborated directly with nonprofit clients to **implement user-centric solutions** that fit their needs

Crcle.ai West Lafayette, IN
Software Engineer (Co-Op until April 2026) October 2025 – Present

- Designed and implemented **Retrieval-Augmented Generation (RAG)** workflows to **enhance Large Language Model (LLM) context accuracy**
- Developed **high-performance mobile interfaces** using React Native, ensuring feature parity and consistent UI/UX across both iOS and Android platforms
- Built **seamless browser integrations** that bridge **user workflows with AI backends**, enabling instant context retrieval and **intelligent overlay features** across web applications

PERSONAL PROJECTS

UPlate

- Engineered an **AI-powered meal planner** by integrating the **Google Gemini Generative AI API** to generate **personalized daily meal plans** tailored to user-calculated macronutrient goals
- **Developed from the ground up using Flutter** to implement a **custom, responsive UI** featuring a stateful, multi-page onboarding flow to seamlessly **collect user biometrics**
- Implemented an offline-first data architecture **using SQLite** for instant app responsiveness and persistence, **with asynchronous data synchronization** to a **Firestore cloud backend**

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- Architected and deployed a dynamic, **full-stack personal portfolio website** to showcase skills and professional experience
- Engineered the frontend with **React and TypeScript**, creating a responsive, mobile-first user interface
- Developed a **robust backend using Node.js** to handle API requests and serve project data, ensuring efficient performance

TECHNICAL SKILLS

Languages: React, TypeScript, Python, Java

Libraries and Tools: RAG, LLMs, Hugging Face, NumPy, SciPy, Matlab, Gazebo, ROS, Node, React, Tailwind

References Available Upon Request